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Dynamic trajectory diffusion model for all-in-one image restoration

Yimin Xu [a,b,*], Yunshan Zhong [c,b], Fei Chao [b,d]

[a] School of Computer Science, Shenyang Aerospace University, Shenyang, Liaoning, China
[b] Media Analytics and Computing Laboratory, Department of Artificial Intelligence, School of Informatics, Xiamen University, Xiamen, Fujian, China
[c] School of Computer Science and Technology, Hainan University, Haikou, Hainan, China
[d] Department of Computer Science Aberystwyth University, Wales, UK

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ABSTRACT

The all-in-one image restoration is an essential capability for practical applications, as it reconstructs various image degradations within a cohesive model framework. Recent advancements in all-in-one image restoration have leveraged diffusion models to map the distribution of diverse degraded images onto a common distribution, aiming to reconstruct latent clean images by simultaneously recovering the task-specific distribution and the degraded residuals. Despite their impressive performance, these methods overlook the intrinsic disparities present in each degraded image, opting to reconstruct clean images with a uniform set of diffusion step sizes, that is, a static diffusion trajectory. To address this limitation, we introduce a novel Dynamic Trajectory Diffusion Model (DT-Diff), which tailors the diffusion trajectory to the characteristics of the input degraded images. Specifically, we integrate a Diffusion Residual Modulator (DRM) and a Diffusion Distribution Modulator (DDM) into the diffusion process, enabling dynamic adjustments to the diffusion trajectory in response to the input images. We also incorporate a lightweight Trajectory Selector (TS) that autonomously identifies the optimal diffusion trajectory during inference. With a negligible increase in computational cost, our proposed DT-Diff surpasses current state-of-the-art all-in-one image restoration methods across five benchmark image restoration datasets. Code is available at https://github.com/xuyimin0926/DT-Diff.

1. Introduction

Image restoration seeks to reconstruct latent clean images by estimating the residual components between the original clean images and their corrupted counterparts. All-in-one image restoration techniques, as opposed to traditional methods, address a multitude of image degradation issues – such as denoising (Liu et al., 2024), deblurring (Mao et al., 2024; Tang et al., 2026), deraining (Wei et al., 2019), and low-light enhancement (Guan et al., 2025; Zhao & Shi, 2025) – within a single, cohesive framework. This unified approach negates the necessity for training multiple discrete models for each specific degradation sample. Moreover, it enhances the flexibility to restore images from a diverse array of unknown corruptions and varying severity levels, thereby providing a more robust and adaptable image restoration process (Wang et al., 2023a).

In contrast to the design of independent models of each image restoration task, all-in-one approaches must encapsulate the distributions of a variety of degradations, which significantly increases the complexity of the restoration process. To more effectively learn the distribution of different degradations, works such as Chen et al. (2021), Li et al. (2020) have adopted a multi-encoder architecture, where each task is handled by its respective encoder. However, the separate input encoders incur the cost of additional parameters. Consequently, some research, as seen in Li et al. (2022), Potlapalli et al. (2024), has concentrated on extracting task-specific features as supplementary prompts to guide the restoration process for images with various types of degradation. While these methods have demonstrated impressive results in reconstructing images with different degradations, they primarily concentrate on exploiting task-specific information, often overlooking the shared information among tasks. Accordingly, Zheng et al. (2024a) have recently introduced a diffusion-based DiffUIR, designed to efficiently harness the shared information. DiffUIR first projects input images into a shared distribution space and then proceeds to recover the underlying clean images. This is achieved by concurrently reconstructing the task-specific distribution and the degraded residual components during the reverse diffusion process.

While DiffUIR successfully restores high-quality clean images by leveraging shared information, it fails to account for the intrinsic

* Corresponding author.
E-mail addresses: xuyimin@sau.edu.cn (Y. Xu), zhongyunshan@hainanu.edu.cn (Y. Zhong), fchao@xmu.edu.cn, fec10@aber.ac.uk (F. Chao).

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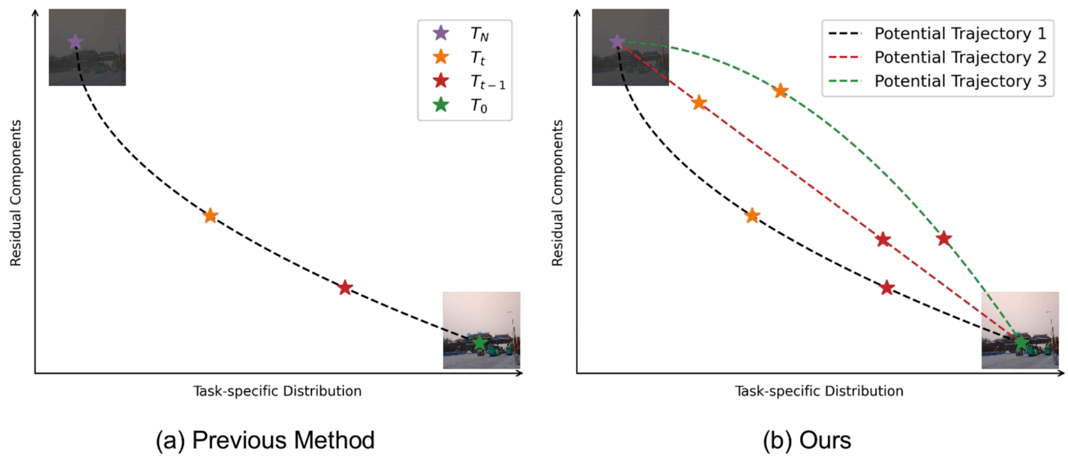

Fig. 1. Illustration of diffusion trajectories. Previous methods follow a static trajectory across all degraded input samples, while our proposed DT-Diff endorses multiple diffusion trajectories, thereby dynamically assigning different potential trajectories to different inputs.

differences among input degraded images. Specifically, envision the reconstruction process of task-specific distributions along the x-axis and that of the residual components along the y-axis. As depicted in Fig. 1(a), by marking the diffusion status at each timestamp as a scatter point, the reverse diffusion process, from degraded to clean images, can be represented as a trajectory by connecting all timestamps sequentially. DiffUIR employs a uniform reverse diffusion trajectory for all input samples, thereby disregarding the profound influence that distinct reverse diffusion trajectories exert on various types of image degradations, which in turn significantly affect the reconstruction outcomes. More broadly, current diffusion-based all-in-one restoration methods (Cheng et al., 2025; Luo et al., 2025) typically adopt a unified reverse diffusion process for all degradation types, potentially restricting their flexibility and task-specific adaptation.

To surmount these limitations, we introduce the Dynamic Trajectory Diffusion Model (DT-Diff), a novel approach that dynamically assigns diffusion trajectories to suit the characteristics of the input degraded images. We introduce a Diffusion Residual Modulator (DRM) and a Diffusion Distribution Modulator (DDM) into the model, serving as two formative components within the diffusion process. These modulators eschew a static diffusion trajectory in favor of a spectrum of potential trajectories with variable velocities, facilitating the recovery of both the degraded residual components and the task-specific distributions. In cases of images with mild degradation, it is conceivable to reconstruct clean images by prioritizing the recovery of residual components before the task-specific distribution, as exemplified by Trajectory 3 in Fig. 1. This approach capitalizes on the shared distribution's capacity to furnish adequate information for predicting degradation residual components. On the flip side, for images with more intricate degradations, recovering the task-specific distribution at the outset affords a more precise estimation of the residual components, thus enhancing the reconstruction process. Moreover, to ensure that each input degraded image is matched with the most fitting diffusion trajectory, we have crafted a lightweight Trajectory Selector (TS). This proposed selector adeptly chooses the optimal combination of DRM and DDM, thereby enabling the reconstruction of latent clean images along the most efficacious diffusion trajectories.

The collective contributions of the paper includes:

- We introduce the Dynamic Trajectory Diffusion Model (DT-Diff) for all-in-one image restoration. Unlike models that rely on a static diffusion trajectory, DT-Diff dynamically tailors trajectories to the degree of degradation present in various input images.
- DT-Diff integrates two diffusion modulator terms — the Diffusion Residual Modulator (DRM) and the Diffusion Distribution Modulator (DDM) — as well as a Trajectory Selector (TS). These modulators provide a range of restoration velocities, while the TS selects the most suitable trajectory for each degraded image.
- Through extensive experimentation across five distinct image restoration tasks, DT-Diff has been demonstrated to achieve superior restoration quality, with an average PSNR improvement of 0.87 over current leading methods, all without incurring significant computational overhead.

2. Related work

2.1. Image restoration

Image restoration is a key area in artificial intelligence, which seeks to restore clean images from damaged ones. Traditionally, researchers have built specific neural networks for each restoration task, like removing rain (Fu et al., 2019; Zhang & Patel, 2018), enhancing low-light images (Li et al., 2024; Yang et al., 2023), or clearing up haze (Dong et al., 2020; Qu et al., 2019). However, these specialized networks don't work well on other tasks. Recently, there's been a move towards creating models that can handle multiple types of image degradation. For example, AirNet (Li et al., 2022) uses a special encoder to understand how images are degraded and then restore them; PromptIR (Potlapalli et al., 2024) uses prompts to adapt to different restoration tasks within a single framework; Painter (Wang et al., 2023b), ProRes (Ma et al., 2023), and DA-CLIP (Luo et al., 2023a) refine large models with prompt learning to improve image restoration. Additionally, some methods like the one by Zheng et al. (2024a) use diffusion models to restore images by mapping them to a common format and then reconstructing the clean image.

2.2. Diffusion model

Diffusion models (Chang et al., 2026; Rombach et al., 2022; Wang et al., 2025, 2026; Zheng et al., 2024b), initially crafted for image generation, employ a Markov Chain to iteratively refine noisy data towards a desired distribution. In a recent development, diffusion models have been adapted for image restoration tasks, aiming to sequentially reconstruct latent clean images. ShadowDiffusion (Guo et al., 2023) pioneers the use of diffusion models for restoring shadow-free images, progressively refining the estimated shadow mask as an auxiliary task for the diffusion generator, which significantly enhances the overall quality of reconstruction. Recognizing that image generation is not solely a generative task, RDDM (Liu et al., 2024) proposes reconstructing clean images by replacing the Gaussian noise input with a degraded image and estimating the corresponding residual degradation components. This

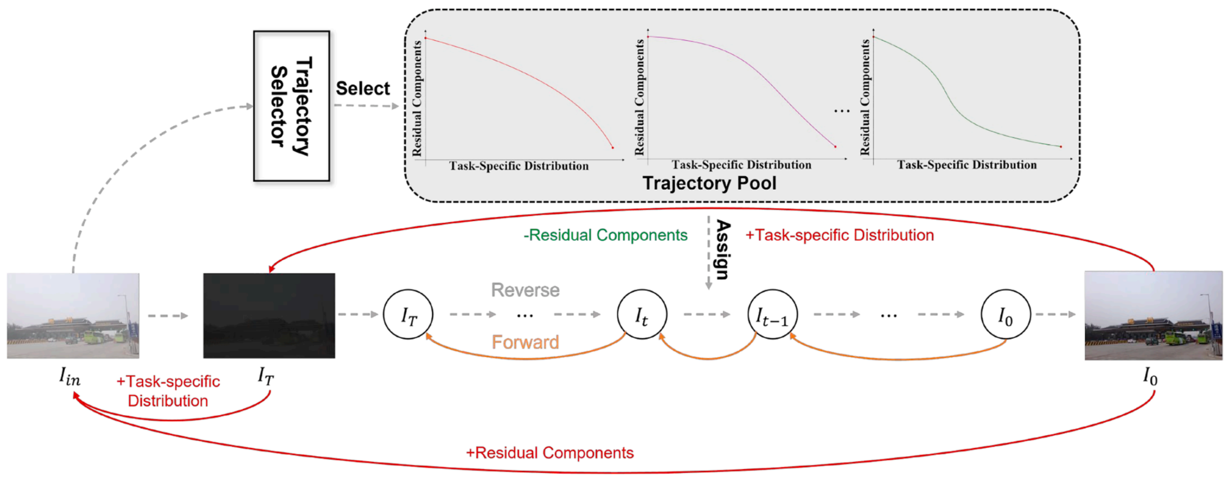

Fig. 2. Overall pipeline of the proposed Dynamic Trajectory Diffusion Models (DT-Diff).

approach has yielded impressive results across several image restoration datasets. DiffUIR (Zheng et al., 2024a) capitalizes on the commonalities among tasks to enhance image restoration performance with a single, unified framework. Despite its restoration capabilities, the universal trajectory employed during the reverse diffusion process does not cater to the specific needs of all input samples. This highlights the importance of selecting the most suitable reverse diffusion trajectories for each distinct input sample.

3. Methodologies

3.1. Problem definition

All-in-one image restoration aims to learn a unified mapping that reconstructs clean images from inputs corrupted by diverse degradation types and severities. Let the degraded image be denoted as $I_{deg} = \mathcal{D}(I_{clean};\theta)$, where $\mathcal{D}(\cdot)$ represents an unknown degradation process parameterized by θ, which may include multiple degradation types (e.g., noise, blur, rain, haze, low-light) and varying severity levels.The objective of all-in-one image restoration is to learn a function:

$$\mathcal{F}_\phi : I_{deg} \rightarrow I_{clean}, \tag{1}$$

that is robust to unknown and mixed degradations without requiring prior knowledge of θ.

3.2. Preliminaries

Image restoration aims to reconstruct the latent clean image I_0 from a degraded input image I_{in}. The input image can be conceptualized as a superposition of the latent clean image I_0 and a residual component I_{res}, such that:

$$I_{in} = I_0 + I_{res}. \tag{2}$$

Denoising diffusion models (Ho et al., 2020) propose to start from a Gaussian noise $I_T \in \mathcal{N}(0, I)$ for generating a clean image I_0 by T sequential time steps of iterative refinement:

$$p_\theta(I_{t-1}|I_t) = \mathcal{N}\left(I_{t-1}; \mu_\theta(I_t, t), \sigma_t I\right), \tag{3}$$

where $t = T \rightarrow 1$; $\mu_\theta(I_t, t)$ and σ_t are the estimated mean and variance, defined as:

$$\mu_\theta(I_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(I_t - \frac{\sqrt{1-\bar{\alpha}_t}}{1-\alpha_t}\epsilon_\theta(I_t, t)\right),$$
$$\sigma_t = \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}, \tag{4}$$

in which $\epsilon_\theta(I, t)$ returns the estimated noise, α controls the velocity of adding noise in the forward diffusion process, and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. Eq. (3) can be reparamaterized as:

$$I_{t-1} = \mu_\theta(I_t, t) + \sigma_t \epsilon, \tag{5}$$

where σ_t denotes variance. The ϵ denotes a random Gaussian noise, usually set to 0 for a deterministic sampling.

In the realm of image restoration, relying solely on Gaussian noise to estimate the underlying clean image often lacks adequate conditional guidance. Therefore, RDDM (Liu et al., 2024) considers integrating the degraded input image I_{in} into the initial point I_T:

$$I_T = I_{in} + \bar{\beta}_T \epsilon, \tag{6}$$

where $\bar{\beta}_t = \sum_{i=1}^{t} \beta_i$ and $\{\beta_i\}_{i=1}^{T}$ is a set of pre-given parameters. Since all-in-one image restoration aims to recover images corrupted by various types of image degradation, merely incorporating the degraded information into I_T doesn't solve the complexity of all-in-one image restoration tasks, as it overlooks the shared information among tasks. Thus, DiffUIR (Zheng et al., 2024a) further integrates the original degraded image I_{in} as:

$$I_T = I_{in} - \bar{\delta}_T I_{in} + \bar{\beta}_T \epsilon. \tag{7}$$

Similar to $\bar{\beta}_t$, $\bar{\delta} = \sum_{i=1}^{t} \delta_i$ defines the velocity of depressing task-specific components in the forward diffusion process. Therefore, with $\bar{\delta}_T$ closer to 1, the reverse diffusion starts from a purer Gaussian distribution. Hence, the reverse diffusion process of DiffUIR can be denoted as the concurrent reconstruction of task-specific distributions and the residual components as Liu et al. (2024):

$$I_{t-1} = \mu_\theta(I_t, I_{res}^{\theta}, I_{in}). \tag{8}$$

Despite considering shared information, the reconstruction of all samples is performed using a universal reverse diffusion trajectory, which overlooks the inherent disparities among input samples. This oversight results in a limited performance increase, underscoring the necessity for a nuanced trajectory assignment strategy for various degraded images.

3.3. Dynamic trajectory diffusion model

The proposed Dynamic Trajectory Diffusion Model (DT-Diff) comprises two primary components: a flexible diffusion process incorporating a Diffusion Distribution Modulator (DDM) and a Diffusion Residual Modulator (DRM), and a Trajectory Selector (TS), as illustrated in Fig. 2. The DDM and DRM endow the diffusion process with multiple potential trajectories, while the TS is designed to select the most appropriate trajectory for each degraded sample.

3.3.1. Flexible diffusion process

The one-step forward diffusion process in DiffUIR, as detailed in the literature (Zheng et al., 2024a), is governed by the following equation:

$$I_t = I_{t-1} + \alpha_t I_{res} + \beta_t \epsilon_{t-1} - \delta_t I_{in}, \tag{9}$$

where α_t regulates the diffusion velocity of the degradation residual components I_{res} and δ_t controls the diffusion velocity of the task-specific distribution I_{in}. The combination of these velocity lists, $[\alpha_1, \dots, \alpha_t]$ and $[\delta_1, \dots, \delta_t]$, delineates a diffusion trajectory from the latent clean image I_0 to the input corrupted images I_{in}.

To facilitate multiple potential trajectories, our DT-Diff incorporates a Diffusion Residual Modulator (DRM) and a Diffusion Distribution Modulator (DDM), offering a variety of velocity options for the degradation residuals and task-specific distributions, respectively. Specifically, the DRM and DDM provide M and N options, enabling mostly $M \times N$ possible diffusion trajectories. At time step t, the DRM with velocity option m is denoted by $\alpha_t^m \in \boldsymbol{\alpha}^m = [\alpha_1^m, \dots, \alpha_{\tilde{T}_m}^m]$, and the DDM with velocity option n is $\delta_t^n \in \boldsymbol{\delta}^n = [\delta_1^n, \dots, \delta_{\tilde{T}_n}^n]$, where T_m and T_n denote the amounts of diffusion steps for the options m and n, respectively. Intuitively, the DRM mainly controls how fast the degradation residuals are accumulated during diffusion, while the DDM regulates how strongly the input image distribution is attenuated. Therefore, selecting different DRM-DDM pairs corresponds to choosing different diffusion velocities and trajectory patterns. The forward diffusion process is modified as:

$$\begin{aligned} I_t &= I_{t-1} + (\alpha_t + \alpha_t^m) I_{res} + \beta_t \epsilon_{t-1} - (\delta_t + \delta_t^n) I_{in} \\ &= I_{t-2} + (\alpha_t + \alpha_t^m + \alpha_{t-1} + \alpha_{t-1}^m) I_{res} \\ &\quad + \sqrt{\beta_t^2 + \beta_{t-1}^2}\,\epsilon_{t-2} - (\delta_t + \delta_t^n + \delta_{t-1} + \delta_{t-1}^n) I_{in} \\ &= I_0 + (\bar{\alpha}_t + \bar{\alpha}_t^m) I_{res} + \bar{\beta}_t \epsilon - (\bar{\delta}_t + \bar{\delta}_t^n) I_{in}. \end{aligned} \tag{10}$$

Therefore, the forward diffusion process can be further expanded to represent the cumulative effect of the DRM and DDM over multiple timesteps, ultimately converging to a simplified form that relates the initial image I_0, the residuals I_{res}, and the input noise ϵ to the final image I_t at the endpoint of the diffusion process. At the endpoint $t = T_m$, $\bar{\alpha}_t + \bar{\alpha}_t^m = 1$, therefore, Eq. (10) is equivalent to:

$$I_{T_M} = \left(1 - (\bar{\delta}_{T_m} + \bar{\delta}_{T_m}^n)\right) I_{in} + \bar{\beta}_{T_m} \epsilon. \tag{11}$$

Specifically, when $M = N = 1$, the trajectory pool collapses into a single configuration, and the modulation coefficients of DRM and DDM become fixed constants across all timesteps. In this case, the modulation no longer introduces adaptive variation, and the reverse diffusion update reduces to a unified form where $\bar{\alpha}_t + \bar{\alpha}_t^n = \bar{\alpha}_t^{fixed}$ and $\bar{\delta}_t + \bar{\delta}_t^n = \bar{\delta}_t^{fixed}$. Therefore, Eq. (10) degenerates to

$$I_t = I_0 + \bar{\alpha}_t^{fixed} I_{res} + \bar{\beta}_t \epsilon - \bar{\delta}_t^{fixed} I_{in}. \tag{12}$$

which is identical to the forward formulation in DiffUIR. As a result, both the reverse diffusion dynamics and the resulting trajectory are equivalent to those of DiffUIR.

3.3.2. Flexible reverse diffusion process

In the reverse diffusion process, the reconstruction of latent clean images involves simultaneously restoring the task-specific distribution and the degraded residual components derived from Eq. (11). iven the forward transition defined in Section 3.3.1, the reverse update can be derived by expressing the posterior of the previous state conditioned on the current state and the observed input. We therefore follow the standard diffusion derivation and start from Bayes' theorem. Following DiffUIR (Zheng et al., 2024a), the posterior distribution, denoted as $p_\theta(I_{t-1}|I_t)$, can be formulated using Bayes' theorem:

$$\begin{aligned} &p_\theta(I_{t-1}|I_t) \to q(I_{t-1}|I_t, I_{in}, I_0^\theta, I_{res}^\theta) \\ &= q(I_t|I_{t-1}, I_{in}, I_{res}^\theta) \frac{q(I_{t-1}|I_0^\theta, I_{res}^\theta, I_{in})}{q(I_t|I_0^\theta, I_{res}^\theta, I_{in})}, \end{aligned} \tag{13}$$

where $q(\cdot)$ symbolizes the probability distribution that emerges from the forward diffusion mechanism. The predicted residual image I_{res}^θ is generated using a U-Net model with parameters θ. When the latent clean image I_0 is inaccessible during the reverse diffusion process, it is approximated by $I_0^\theta = I_{in} - I_{res}^\theta$. To be more specific, the q in Eq. (13) can be denoted as:

$$\begin{aligned} q(I_t|I_{t-1}, I_{in}, I_{res}^\theta) &\sim \mathcal{N}(I_{t-1} + (\alpha_t + \alpha_t^m) I_{res}^\theta - (\delta_t + \delta_t^n) I_{in}, \beta_t^2 I) \\ q(I_{t-1}|I_0^\theta, I_{res}^\theta, I_{in}) &\sim \mathcal{N}(I_0 + (\bar{\alpha}_{t-1} + \bar{\alpha}_{t-1}^m) I_{res}^\theta - (\bar{\delta}_{t-1} + \bar{\delta}_{t-1}^n) I_{in}, \bar{\beta}_{t-1}^2 I) \\ q(I_t|I_0^\theta, I_{res}^\theta, I_{in}) &\sim \mathcal{N}(I_0 + (\bar{\alpha}_t + \bar{\alpha}_t^m) I_{res}^\theta - (\bar{\delta}_t + \bar{\delta}_t^n) I_{in}, \bar{\beta}_{t-1}^2 I) \end{aligned} \tag{14}$$

Considering the Gaussian nature of the probability distribution in the forward diffusion, Eq. (13) can be further elaborated to select the mth DRM and nth DDM as:

$$\begin{aligned} p_\theta^{m,n}(I_{t-1}|I_t) &\sim \exp[-\frac{1}{2}\left(\left(\frac{\bar{\beta}_t^2}{\beta_t^2 \bar{\beta}_{t-1}^2}\right) I_{t-1}^2\right. \\ &- 2\left(\frac{I_t + (\delta_t + \delta_t^n) I_{in} - (\alpha_t + \alpha_t^m) I_{res}^\theta}{\beta_t^2}\right. \\ &+ \left.\left.\frac{I_0^\theta + (\bar{\alpha}_{t-1} + \bar{\alpha}_{t-1}^m) I_{res}^\theta - (\bar{\delta}_{t-1} + \bar{\delta}_{t-1}^n) I_{in}}{\bar{\beta}_{t-1}^2}\right) I_{t-1}\right) \\ &+ C(I_t, I_0^\theta, I_{res}^\theta, I_{in})]. \end{aligned} \tag{15}$$

The mean $\mu_\theta^{m,n}$ and variance $\sigma_\theta^{m,n}$ at timestep t become:

$$\begin{aligned} \mu_\theta^{m,n}(I_t, t) &= I_t - (\alpha_t + \alpha_t^m) I_{res}^\theta + (\delta_t + \delta_t^n) I_{in} - \frac{\beta_t^2}{\bar{\beta}_t} \epsilon^\theta, \\ \sigma_\theta^{m,n}(I_t, t) &= \frac{\beta_t^2 \bar{\beta}_{t-1}^2}{\bar{\beta}_t^2}. \end{aligned} \tag{16}$$

in which ϵ^θ represents the estimated noise at timestep t, which can be inferred using Eq. (10) based on the estimated residual component I_{res}^θ. However, due to the deterministic nature of image restoration tasks, we adhere DiffUIR (Zheng et al., 2024a), which advocates for the exclusion of stochastic noise. Consequently, the I_{t-1} can be derived with reparameterization trick as:

$$I_{t-1} = I_t - (\alpha_t + \alpha_t^m) I_{res}^\theta + (\delta_t + \delta_t^n) I_{in}. \tag{17}$$

3.3.3. Trajectory selector

In the context of a diffusion model offering a multitude of trajectories, we introduce a trajectory selector designed to ascertain the most suitable diffusion pathway for each impaired sample. Initially, the degraded images undergo a sequence of two successive residual blocks, as delineated by He et al. (2016), to distill the pertinent input features f_{in}, which is articulated as follows:

$$f_{in} = R_2 \circ R_1(I_{in}), \tag{18}$$

where R_1 and R_2 denote the residual blocks. Subsequently, these captured features are conveyed to an average pooling layer, followed by two linear layers L_1 and L_2, as follows:

$$logits = L_2 \circ L_1(AvgPool(f_{in})), \tag{19}$$

where $logits \in \mathbb{R}^{M+N}$ signifies the output logits of the input image, wherein the initial M logits, specially $logits[1 : M]$, are aligned with the DRM's logits, while $logits[M+1 : M+N]$ symbolize the logits pertaining to the DDM.

3.4. Training objective

The training process of the proposed DT-Diff can be bifurcate into two distinct phases: the training of the flexible diffusion model and the training of the trajectory selector. Initially, the focus is on the diffusion model, where at each timestep t in the forward diffusion process, a trajectory is randomly chosen, characterized by the endpoints t_{T_n} and t_{T_m} of

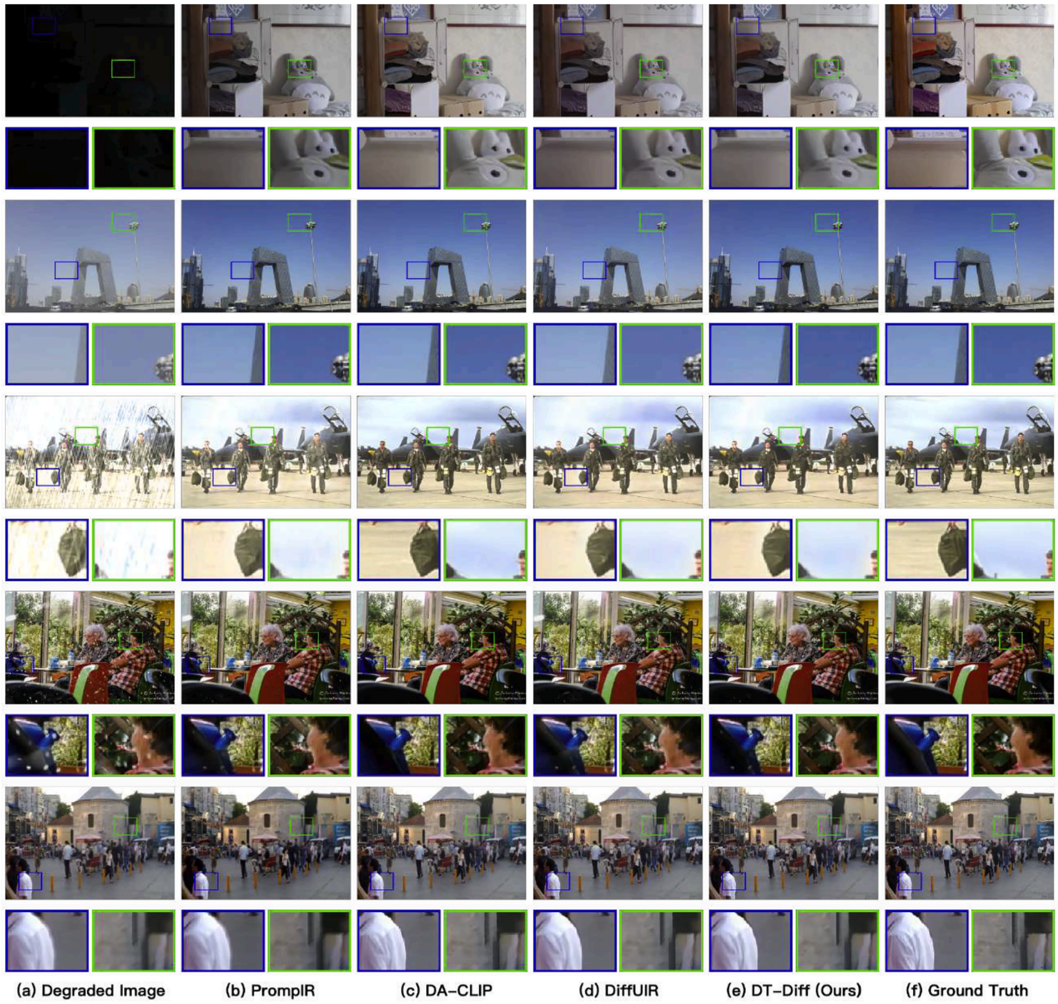

Fig. 3. Visualization of our DT-Diff and existing methods including PromptIR (Potlapalli et al., 2024), DA-CLIP (Luo et al., 2023a), and DiffUIR (Zheng et al., 2024a). Rows 1 through 5 sequentially illustrate the performance of these models in the restoration of images affected by low-light conditions, haze, rain, snow, and blurring. Zoom in for the best view.

the DRM and DDM, respectively, satisfying the condition $t_{T_n} \geq t_{T_m} \geq t$. The diffusion model is then tasked with forecasting the degraded residual components from state I_t, as per Eq. (10). The training objective of the diffusion model is formulated as an L1 loss, which quantifies the discrepancy between the estimated degraded residual components $I^{\theta}_{res}(I_t, t)$ and their true values:

$$\mathcal{L} = \mathbb{E}_{t,I_t,I_{res}}[||I_{res} - I^{\theta}_{res}(I_{in}, I^{m,n}_t, t)||_1], \tag{20}$$

where $I^{m,n}_t$ denotes the diffusion status at timestamp t with mth DRM and nth DDM. Therefore, the pre-trained diffusion model can support different diffusion schedules within a unified model. With a pre-trained diffusion model in place, the trajectory selector is subsequently trained to enable adaptive trajectory selection tailored to diverse input samples. The loss function for this phase is constructed as a weighted L1-norm, which measures the difference between the reconstructed images and the actual clean images:

$$\mathcal{L}_{ts} = \sum_m \sum_n p_m \cdot p_n \cdot ||I_0 - \tilde{I}^{\theta}_0||_1, \tag{21}$$

where p_m and p_n represent the probabilities of selecting the mth DRM and nth DDM, respectively. These probabilities are derived by applying the softmax function to the output logits produced by Eq. (19). The term $\tilde{I}^{\theta}_0$ denotes the estimated clean image obtained through DDIM sampling during the reverse diffusion process. The training procedure can be shown in Algorithms 1 and 2.

Algorithm 1 Training procedure of flexible diffusion model.

Require: Total training iterations T
1: Initialize diffusion model parameters θ
2: Initialize trajectory selector parameters ϕ
3: Freeze the weight ϕ of trajectory selector
4: **for** t in $[0, T]$ **do**
5: Randomly sample a DRM and a DDM from the set of all available choices
6: Perform the forward diffusion process according to Eq. (10)
7: Compute the diffusion loss using Eq. (20) and update the diffusion model parameters θ via back propagation
8: **end for**

4. Experiments

4.1. Experimental settings

4.1.1. Datasets

In alignment with the settings in DiffUIR (Zheng et al., 2024a), we have undertaken experiments across five distinct image restoration tasks: image deraining, image dehazing, low-light image enhancement, image desnowing, and image deblurring. For the image deraining, we have employed a comprehensive dataset as referenced in Jiang et al.

Table 1
Comparison of our DT-Diff over task-specific image restoration methods and all-in-one image restoration methods. The FLOPs of each method are calculated with 256×256 image patches. The best and second best are marked in **red** and *blue*.

Method	Deraining		Enhancment		Desnowing		Dehazing		Deblurring		Average		Complexity	
	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	Params (M)	FLOPS (G)
Task-specific Image Restoration Methods														
SwinIR (**ICCV'21**)	–	–	17.81	0.723	–	–	21.50	0.891	24.52	0.773	–	–	0.9	752.13
MIRNet-v2 (**TPAMI'22**)	–	–	24.74	0.851	–	–	24.03	0.927	26.30	0.799	–	–	5.9	140.92
DehazeFormer (**TIP'22**)	–	–	–	–	–	–	34.29	0.983	–	–	–	–	4.63	48.64
Restormer (**TIP'23**)	33.96	0.935	20.41	0.806	–	–	30.87	0.969	32.92	0.961	–	–	26.12	141.00
MAXIM (**CVPR'22**)	33.24	0.933	23.43	0.863	–	–	34.19	0.985	32.86	0.940	–	–	14.10	216.00
DRSFormer (**CVPR'23**)	33.15	0.927	–	–	–	–	–	–	–	–	–	–	33.7	242.90
IR-SDE (**ICML'23**)	–	–	20.45	0.787	–	–	–	–	30.70	0.901	–	–	34.2	98.3
WeatherDiff (**TPAMI'23**)	–	–	–	–	33.51	0.939	–	–	–	–	–	–	82.96	–
RDDM (**CVPR'24**)	30.74	0.903	23.22	0.899	32.55	0.927	30.78	0.953	29.53	0.876	–	–	36.26	9.88
All-in-one Image Restoration Methods														
Restormer (**CVPR'22**)	27.10	0.843	17.13	0.542	28.61	0.876	22.79	0.706	26.36	0.814	23.60	0.756	26.12	141.00
AirNet (**CVPR'22**)	24.87	0.773	14.83	0.767	27.63	0.860	25.47	0.923	26.92	0.811	23.94	0.849	8.93	30.13
Painter (**CVPR'23**)	29.49	0.868	22.40	0.872	–	–	–	–	–	–	–	–	307	248.9
IDR (**CVPR'23**)	–	–	21.34	0.826	–	–	25.24	0.943	27.87	0.846	–	–	15.34	–
ProRes (**ArXiv'23**)	30.67	0.891	22.73	0.877	–	–	–	–	27.53	0.851	–	–	307	248.9
PromptIR (**NIPS'23**)	29.56	0.888	22.89	0.847	31.98	0.924	32.02	0.952	27.21	0.817	28.15	0.874	35.59	15.81
MPerceiver+ (**CVPR'24**)	30.89	0.901	22.49	0.854	31.11	0.918	21.08	0.651	**32.98**	**0.961**	27.71	0.857	–	–
DA-CLIP (**ICLR'24**)	28.96	0.853	24.17	0.882	30.80	0.888	31.39	0.983	25.39	0.805	28.63	0.871	174.1	118.5
DiffUIR-L (**CVPR'24**)	*31.03*	*0.904*	*25.12*	*0.907*	*32.65*	*0.927*	*32.94*	*0.956*	*29.17*	*0.864*	*30.18*	*0.912*	36.26	9.88
DT-Diff (Ours)	**32.87**	**0.925**	**25.26**	**0.908**	**32.69**	**0.928**	**35.54**	**0.994**	28.90	0.858	**31.05**	**0.923**	36.55	10.00

Algorithm 2 Training procedure of trajectory selector.

Require: Total training iterations T', M predefined DRMs and N predefined DDMs
Load pretrained parameters θ and ϕ from Algorithm 1
Freeze the weight θ of Flexible Diffusion Model
for t in $[0, T']$ **do**
  for $m = 1$ to M **do**
    for $n = 1$ to N **do**
      Perform 3-step DDIM sampling using trajectory parameterized (m, n)
    end for
  end for
  Compute selector loss $\mathcal{L}_{ts}$ from Eq. (21)
  Update selector parameters ϕ
end for

(2020), Wang et al. (2022), encompassing 13,712 images for training and 4298 images for testing. For the domain of image dehazing, we have selected the RESIDE dataset (Li et al., 2018), comprising 313,950 hazy outdoor image pairs for training and 500 pairs for testing. For low-light image enhancement, the renowned LOL dataset (Wei et al., 2018) has been utilized, offering 485 training pairs and 15 pairs for testing. In the case of image desnowing, the Snow100K dataset (Liu et al., 2018), which contains 50,000 unique snowy images for both training and testing, has been leveraged, and we have conducted evaluations on Snow100K-S and Snow100K-L to assess the performance of DT-Diff on images with varying snowflake sizes. Lastly, for image deblurring, the GoPro dataset (Nah et al., 2017), featuring 2103 training samples and 1111 testing samples, has been utilized. If not otherwise specified, all experiments, including testing and ablation studies, are conducted on the aforementioned five datasets. Besides these synthetic datasets, we also evaluate the performance of DT-Diff on several real-world datasets, including the Practical (Yang et al., 2017) dataset for image deraining and the MEF (Ma et al., 2015), NPE (Wang et al., 2013), and DICM (Lee et al., 2013) dataset for low-light enhancement.

4.1.2. Evaluation metrics

To ensure a rigorous comparison between our proposed DT-Diff and existing benchmark image restoration methods, we have employed the Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index Measure (SSIM). Beyond the performance metrics of each individual task, we also present the average PSNR/SSIM among all tasks, recognizing that the simultaneous presence of all task types is often impractical in real-world scenarios. Additionally, we report on the computational complexity of each restoration model, with a focus on the number of parameters and the floating point operations (FLOPs) required.

4.1.3. Implementation details

We train DT-Diff on 4 NVIDIA GeForce RTX 3090 GPUs. During training the flexible diffusion model, we utilize the Adam optimizer, setting the learning rate set to 8×10^{-5} and completing a total of 300,000 iterations. To facilitate a multitude of potential diffusion trajectories, 4 DRMs, and 4 DDMs are integrated. We adopt the Unet mentioned in Zheng et al. (2024a) as the diffusion backbone. To train the Trajectory Selector (TS), an additional 5000 iterations are performed with a learning rate of 0.01 and a batch size of 10. During this phase, each candidate diffusion trajectory has 3 sampling steps derived by DDIM sampling strategy, and ϵ is set to 0 to enforce deterministic reverse updates. Notably, to ensure a fair comparison with previous image restoration methods, we set ϵ in Eq. (10) to zero, as no noise is introduced in the input images.

4.2. Comparison with the state-of-the-arts

To assess the efficacy of our DT-Diff, we have juxtaposed it with a variety of both task-specific and all-in-one image restoration methods. The task-specific methods include SwinIR (Liang et al., 2021), MIRNet-v2 (Zamir et al., 2022b), DehazeFormer (Song et al., 2023), MAXIM (Tu et al., 2022), DRSFormer (Chen et al., 2023), IR-SDR (Luo et al., 2023b), WeatherDiff (Özdenizci & Legenstein, 2023), and RDDM (Liu et al., 2024). The all-in-one methods encompass Restormer (Zamir et al., 2022a), AirNet (Li et al., 2022), Painter (Wang et al., 2023b), IDR

Table 2
Ablation study evaluating the contribution of DRM, DDM, and TS across five restoration tasks.

Method				Deraining		Enhancment		Desnowing		Dehazing		Deblurring		Average	
Diffusion	DRM	DDM	TS	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
✓				31.03	0.904	25.12	0.907	32.65	0.927	32.94	0.956	29.17	0.864	30.18	0.912
✓	✓		✓	32.16	0.920	25.18	0.907	32.56	0.926	34.15	0.977	28.78	0.857	30.41	0.914
✓		✓	✓	31.52	0.913	25.25	0.908	32.67	0.928	34.80	0.981	28.66	0.853	30.58	0.917
✓	✓	✓	✓	**32.87**	**0.925**	**25.26**	**0.908**	**32.69**	**0.928**	**35.54**	**0.994**	**28.90**	**0.858**	**31.05**	**0.923**

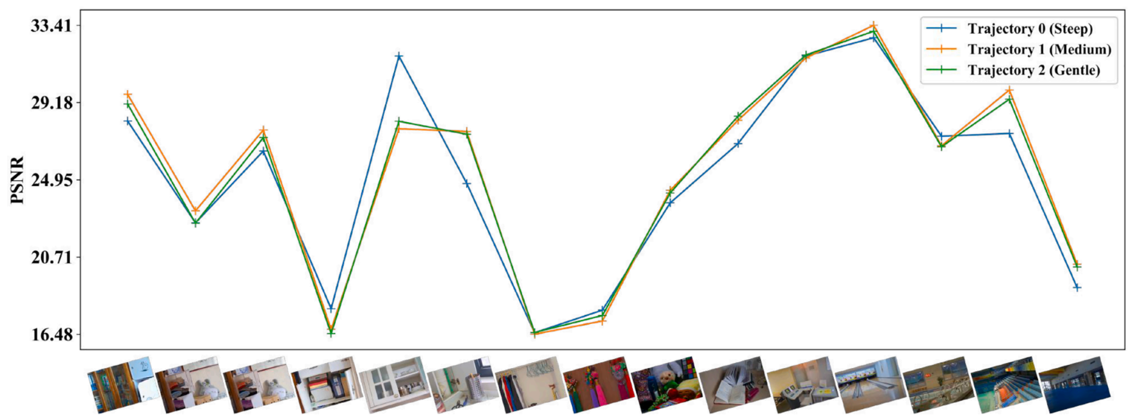

Fig. 4. Per-image PSNR comparison of three trajectory configurations, showing that no single trajectory consistently performs best across all samples, thereby motivating sample-adaptive selection.

(Park et al., 2023), ProRes (Ma et al., 2023), PromptIR (Potlapalli et al., 2024), MPerceiver+ (Ai et al., 2024), DA-CLIP (Luo et al., 2023a) and DiffUIR (Zheng et al., 2024a).

The comparative results in Table 1 indicate DT-Diff generally surpasses DiffUIR in most image restoration tasks. Notably, DT-Diff achieves a higher PSNR of 0.87, outperforming DiffUIR. This model also excels in specific tasks such as image deraining and dehazing without incurring additional computational costs. The visual demonstrations in Fig. 3 further highlight the model's exceptional restoration capabilities across diverse scenarios. Specifically, DT-Diff delivers images of enhanced quality capabilities across diverse scenarios. Specifically, DT-Diff delivers images of enhanced quality and greater detail compared to DiffUIR. This improvement is credited to DT-Diff's capacity to discern and select the optimal restoration trajectory for each sample, thereby streamlining the restoration process.

4.3. Ablation studies

4.3.1. Component efficacy

The efficacy of the proposed DT-Diff is primarily attributed to the sophistication of its flexible diffusion process, which integrates the Diffusion Residual Modulator (DRM), the Diffusion Distribution Modulator (DDM), and the Trajectory Selector (TS). To ensure a fair and comprehensive comparison, the TS is utilized across all possible combinations of DRM and DDM to identify the most suitable trajectory for each sample. As illustrated in Table 2, both the DRM and DDM, when applied independently, significantly improve the performance of image restoration tasks over conventional diffusion models. However, the synergistic effect of DRM and DDM, when combined, results in a further enhancement of performance.

4.3.2. Effectiveness of the TS

In DT-Diff, the TS is tasked with choosing from various trajectories that are individually tailored to different samples. To substantiate the effectiveness of the TS, we conduct a comparative analysis against both random and task-wise assignment strategies, as delineated by Table 3. The TS, by thoroughly considering the unique challenges presented by each sample, markedly outperforms the alternative strategies, demonstrating a significant advantage over both random and task-wise assignments.

Table 3
Comparison between the proposed Trajectory Selector and other selection strategies to demonstrate the efficacy of the TS.

Sample Method	PSNR	SSIM
Random Assignment	29.35	0.901
Task-wise Assignment	30.31	0.914
Trajectory Selector	**31.05**	**0.923**

Table 4
Quantitative results on the UDC benchmark (Toled and Poled subsets), demonstrating that DT-Diff generalizes effectively to real-world underwater degradations.

Method	Toled			Poled		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
DL	21.23	0.656	0.433	13.92	0.449	0.756
IDR	27.91	0.795	0.312	**16.71**	0.497	0.716
PromptIR	16.70	0.688	0.422	13.16	**0.583**	0.619
DiffUIR	29.55	0.887	0.281	15.62	0.424	0.505
DT-Diff	**29.71**	**0.892**	**0.269**	15.70	0.430	**0.497**

Table 5
Results on real-world deraining (Practical) and low-light enhancement (MEF, NPE, and DICM) datasets.

Method	Deraining		Enhancement	
	NIQE ↓	LOE ↓	NIQE ↓	LOE ↓
AirNet	3.55	145.3	3.45	593.13
PromptIR	3.52	28.53	3.31	255.13
DiffUIR	3.38	24.83	3.14	193.43
DT-Diff	**3.29**	**24.65**	**3.08**	**187.38**

Table 6
Comparison of perceptual quality (LPIPS) and running time, showing that DT-Diff achieves consistently lower perceptual error than DiffUIR without increasing inference cost.

Method ↓	Deraining	Enhancement	Desnowing	Dehazing	Deblurring	Running Time (s)
MPerceiver+ (reported)	–	–	–	0.422	0.087	–
DiffUIR	0.0932	0.1674	0.1124	0.0897	0.2351	**0.371**
DT-Diff	**0.0665**	**0.1637**	**0.1050**	**0.0372**	**0.2312**	0.372

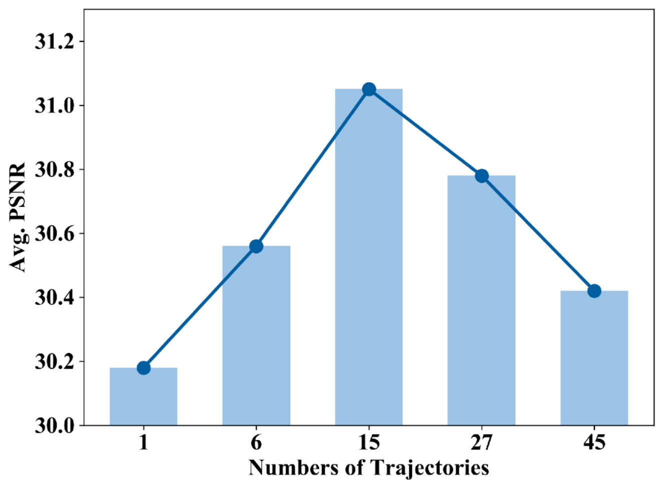

Fig. 5. Relationship between the number of trajectory configurations and average PSNR, demonstrating the existence of an optimal trajectory diversity level rather than monotonic improvement.

4.3.3. Analysis of diffusion trajectories

As depicted in Fig. 4, three unique diffusion trajectories have been randomly selected to evaluate their performance in the context of low-light image enhancement. The trajectories exhibit a progression from steep initial gradients, which **prioritize the restoration of residual components** (e.g. potential trajectory 1 in Fig. 1), to gentler slopes that focus on **task-specific components** (e.g. potential trajectory 3 in Fig. 1), transitioning from Trajectory 0 to Trajectory 2. Notably, Trajectory 0 has proven to be exceptionally effective for restoring images characterized by uniform color distributions, as illustrated by the example restoration provided in Fig. 4. This observation supports the notion that employing diverse trajectories to accommodate varying input samples is an effective strategy. As Fig. 4 presents an illustrative example using the low-light enhancement task to visualize the effect of different trajectories, the proposed trajectory selection mechanism is not limited to this specific task. Its generality is further supported by the quantitative results across multiple restoration tasks, as shown in Table 2, where the use of trajectory selection consistently improves performance under different degradation types.

4.3.4. Experiments on UDC dataset

We have included the results for the UDC dataset. As shown in Table 4, DT-Diff consistently achieves the best or second-best performance across all metrics on both Toled and Poled subsets. The improvements over DiffUIR indicate that dynamically adjusting diffusion trajectories enables the model to better handle complex sensor-induced distortions, which typically involve mixed noise, blur, and exposure inconsistencies.

4.3.5. Experiments on real-world dataset

We further evaluate DT-Diff on real-world deraining and low-light enhancement datasets, where distortion types and severities cannot be precisely modeled. As reported in Table 5, DT-Diff achieves the lowest NIQE and LOE scores, outperforming AirNet, PromptIR, and DiffUIR across both tasks, demonstrating the proposed trajectory modulation enhances the model's ability to recover natural distortions.

4.3.6. Number of DRM and DDM

The quantity of DRMs and DDMs is pivotal in gauging the performance of DT-Diff. Increasing the number of DRM and DDM configurations enlarges the trajectory pool and enhances the model's expressive capacity to handle diverse degradation patterns, enabling finer control over the reverse diffusion process. However, an excessive number of trajectories also increases the complexity of trajectory selection and optimization, introducing redundancy and potentially reducing decision stability. As depicted in Fig. 5, we have determined that the optimal number of potential trajectories is 15, that is, with $M = N = 5$, to achieve peak performance. While $M = N = 5$ the combination of DRMs and DDMs theoretically allows for a maximum of 25 possible trajectories; however, the condition $t_{T_n} \geq t_{T_m}$ reduces this number to 15.

4.3.7. Other quantitative comparsion

As shown in Table 6, we report the perceptual score and the average running time of the proposed DT-Diff with the DiffUIR. Due to the limited availability of LPIPS results and official implementations for other all-in-one methods, comparison is conducted with DiffUIR under unified settings to ensure reproducibility. From Table 6, the proposed DT-Diff reaches higher LPIPS scores, which means the proposed DT-Diff can also generate perceptually better images than the SOTA method. Meanwhile, DT-Diff introduces merely millisecond-level negligible overhead in running time compared with DiffUIR. To ensure completeness, we also include the LPIPS of MPerciever as a reference, however, these results are obtained under different settings and are not directly comparable.

4.4. Discussion

4.4.1. Difference between DT-Diff and adaptive time-step diffusion models

Since adaptive timestep diffusion models are closely related to our method, we provide a detailed comparison between such approaches and DT-Diff. Existing adaptive timestep methods retain the same reverse diffusion procedure and primarily adjust the sampling density of timesteps or the arrangement of noise coefficients within a shared diffusion process. In contrast, DT-Diff introduces multiple reverse diffusion update configurations through the Diffusion Residual Modulator (DRM) and Diffusion Distribution Modulator (DDM). Moreover, prior sample-wise diffusion step adjustment methods typically modify only a single scheduling dimension (e.g., timestep density or noise coefficients). DT-Diff, however, jointly modulates both the residual term and the distribution term via DRM and DDM, enabling more flexible and task-adaptive restoration trajectories.

4.4.2. Relation to DiffUIR

DiffUIR adopts a unified reverse diffusion trajectory, where all input samples follow a shared update schedule during inference. While such a unified strategy simplifies the restoration process, it applies a shared reverse trajectory across different degradation types.

In contrast, DT-Diff expands the reverse diffusion process from a single fixed update trajectory to a configurable family of update trajectories. By introducing the Diffusion Residual Modulator (DRM) and Diffusion Distribution Modulator (DDM), multiple reverse diffusion configurations are constructed to form a trajectory pool. Each configuration corresponds to a distinct balance between deterministic residual correction and stochastic distribution alignment across timesteps.

Therefore, DT-Diff does not merely adjust diffusion step numbers or noise schedules. Instead, it enables trajectory-level flexibility by allowing different input samples to follow structurally different reverse diffusion trajectories, thereby extending the expressive capacity of the reverse process beyond a single predefined update rule. Moreover, when there is only 1 DRM and 1 DDM, the DT-Diff becomes identical to DiffUIR.

5. Limitation

Despite the promising outcomes exhibited by the Dynamic Trajectory Diffusion Model (DT-Diff) across a range of image restoration tasks, the model's adaptability is somewhat constrained by the predefined settings of the Diffusion Residual Modulator (DRM) and Diffusion Distribution Modulator (DDM). For example, the deblurring performance of DT-Diff, as delineated in Table 1, does not achieve the same level of effectiveness as observed with DiffUIR. This discrepancy may stem from the limited selection of available trajectory options within the DT-Diff framework. Additionally, the number of sampling steps in the DT-Diff could be reduced to accelerate the overall inference speed. This aspect warrants further investigation to optimize performance in future studies.

6. Conclusion

In this paper, we have introduced the Dynamic Trajectory Diffusion Model (DT-Diff), a novel approach that significantly advances all-in-one image restoration. This model dynamically customizes diffusion trajectories according to the unique attributes of each degraded image, thereby enhancing the restoration process. By integrating the Diffusion Residual Modulator (DRM) and the Diffusion Distribution Modulator (DDM), along with a Trajectory Selector (TS), DT-Diff has demonstrated superior performance over current state-of-the-art methods across a variety of tasks, including deraining, dehazing, low-light image enhancement, desnowing, and deblurring. Our future work will focus on developing more adaptable diffusion models that do not rely on a predetermined number of DRMs and DDMs. This innovation aims to further improve the model's capability to adeptly handle diverse all-in-one image restoration tasks.

CRediT authorship contribution statement

Yimin Xu: Conceptualization, Data curation, Investigation, Methodology, Software, Writing – original draft; **Yunshan Zhong:** Visualization; **Fei Chao:** Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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